EE-3TR4 Mid Term Exam 2021

Duration of examination: 1 hour
Answer all the questions.
Total marks = 20.

1. The impulse response of a filter is
   $h(t) = 4rect[5(t-2)]$,
   and its input is
   $x(t) = 3\delta(t+2) + 2rect(5t)$. $+ 5\cos(8\pi t)$
   Find its output $y(t)$.
   Note: Use the information on the formula sheet.

Q1 2023

7 marks

2. Consider an AM wave for the following message signal:
   $$m(t) = 2\cos(20\pi t) + 5\sin(40\pi t).$$
   The AM wave is
   $$s(t) = [1 + k_a m(t)]\cos(2000\pi t).$$
   % modulation = 40%. Sketch the magnitude spectrum ($|S(f)|$) of the AM wave. What is the ratio of the power in the sidebands to the total power of the AM wave?

7 marks

3 Explain the principle of operation of envelope detector, and
(a) if the message signal is $m(t) = \cos(2\pi f_m t)$, and the time constant of the output circuit, $R_L C = 1/f_c$, where f_c is the carrier frequency, sketch the output of the envelope detector (qualitatively). Assume $f_c >> f_m$.
(b) if an DSB-SC signal with the message signal $m(t) = \cos(2\pi f_m t)$ is applied to an ideal envelope detector, sketch the output of the envelope detector (qualitatively). Can the message signal be retrieved without distortion? Explain. Assume $f_c >> f_m$.

6 marks

Note: The ideal envelope detector detects the positive side of the envelope without any distortion such as ripple.

Mark distribution: Principle of operation explanation – 2 *marks*
3a - 2 *marks*
3b - 2 *marks*
Total - 6 *marks*

Useful Information

Q 2. Consider an AM wave for the following message signal:

$$m(t) = 2\cos(20\pi t) + 5\sin(40\pi t).$$

The AM wave is

$$s(t) = [1 + k_a m(t)]\cos(2000\pi t).$$

% modulation = 40%. Sketch the magnitude spectrum ($|S(f)|$) of the AM wave. What is the ratio of the power in the sidebands to the total power of the AM wave?

a) b)

$* \cos(2\theta) = 1 - 2\sin^2\theta$

a) $m(t) = 2\cos(20\pi t) + 5\sin(40\pi t)$, $f_1 = 10$, $f_2 = 20$

$s(t) = [1 + k_a m(t)]\cos(2000\pi t)$, $f_c = 1000$

% mod = 40%

envelope distortion:

freq:

$f_c >> W$

$1000 >> 10 = f_1$

$1000 >> 20 = f_2$

time:

$\% \text{ mod} = 100\, k_a \underbrace{\max|m(t)|}_{?}$

$\max[m(t)]$: $0 = m'(t)$

$= [2\cos(20\pi t) + 5\sin(40\pi t)]\ dt$

$= 20\pi \cdot 2 \cdot -\sin(20\pi t) + 40\pi \cdot 5 \cdot \cos(40\pi t)$

$0 = -40\pi\sin(20\pi t) + 200\pi\cos(40\pi t)$

$40\pi\sin(20\pi t) = 200\pi\cos(40\pi t)$

$\sin(20\pi t) = 5\cos(40\pi t)$ $\theta = 20\pi t$, $2\theta = 40\pi t$

$\sin(\theta) = 5\cos(2\theta)$ $\cos(2\theta) = 1 - 2\sin^2\theta$

$= 5[1 - 2\sin^2\theta]$

$\sin\theta = 5 - 10\sin^2\theta$

$0 = 10\sin^2\theta + \sin\theta - 5$

$0 = 10x^2 + x - 5$, $x = \sin\theta$

$x = \frac{-1 \pm \sqrt{1^2 - 4(10)(-5)}}{2(10)}$

$= \frac{-1 \pm \sqrt{201}}{20}$

$= -0.76$, 0.66

$x = \sin\theta = \sin(20\pi t)$

$\theta = \sin^{-1} x = -0.86$, 0.72

$\theta = 20\pi t$

$t = \theta/20\pi$

$= -0.014$, 0.011

$m(t = -0.014) =$

$(t = 0.011) = 6.45$

$m(t) = 2\cos(\theta) + 5\sin(2\theta)$

$m(t = 0.72) = 6.45$

$m(t = -0.86) = -3.64$

$\max|m(t)| = 6.45$

$\% \text{ mod} = 100 \cdot k_a \cdot \max|m(t)|$

$40 = 100 \cdot k_a \cdot 6.45$

$k_a = \frac{40}{645}$

$= 8/129$

$k_a = 0.062$

$s(t) = [1 + k_a m(t)]\cos(2000\pi t)$

$= [1 + k_a(2\cos(20\pi t) + 5\sin(40\pi t))]\cos(2000\pi t)$

$= \underbrace{\cos(2000\pi t)}_{S_1(t)} + \underbrace{k_a \cdot 2\cos(20\pi t)\cos(2000\pi t)}_{S_2(t)} + \underbrace{k_a \cdot 5\sin(40\pi t)\cos(2000\pi t)}_{S_3(t)}$

$= S_1(t) + S_2(t) + S_3(t)$

$S_2(t) = 2k_a\cos(20\pi t)\cos(2000\pi t)$ * trig ID.

$= k_a[\cos(20\pi t - 2000\pi t) + \cos(20\pi t + 2000\pi t)]$

$= k_a[\cos(2\pi t \cdot (10 - 1000)) + \cos(2\pi t(10 + 1000))]$

$S_3(t) = 5k_a\sin(40\pi t)\cos(2000\pi t)$

$= \frac{5}{2} 2k_a\sin(40\pi t)\cos(2000\pi t)$

$= \frac{5}{2}k_a[\sin(2\pi t(20 - 1000)) + \sin(2\pi t(20 + 1000))]$

$S_1(t) = \cos(2\pi 1000 t)$

$S_1(f) = \frac{1}{2}[\delta(f - 1000) + \delta(f + 1000)]$

$S_2(t) = k_a\cos(2\pi\overset{-990}{(10 - 1000)}t) + k_a\cos(2\pi\overset{1010}{(10 + 1000)}t)$

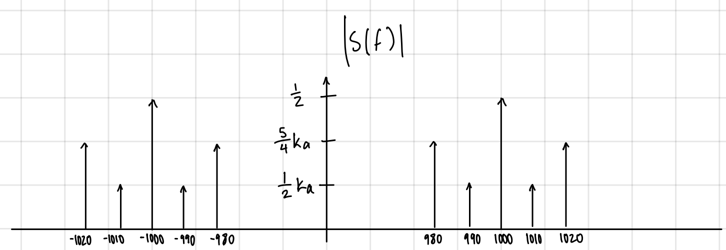
$S_2(f) = k_a\frac{1}{2}[\delta(f + 990) + \delta(f - 990)] + k_a\frac{1}{2}[\delta(f - 1010) + \delta(f + 1010)]$

$S_3(t) = \frac{5}{2}k_a\sin(2\pi\overset{-980}{(20 - 1000)}t) + \frac{5}{2}k_a\sin(2\pi\overset{1020}{(20 + 1000)}t)$

$S_3(f) = \frac{5}{2}k_a\frac{1}{2j}[\delta(f + 980) - \delta(f - 980)] + \frac{5}{2}k_a\frac{1}{2j}[\delta(f - 1020) - \delta(f + 1020)]$

$= \frac{5}{4j}k_a[\delta(f + 980) - \delta(f - 980)] + \frac{5}{4j}k_a[\delta(f - 1020) - \delta(f + 1020)]$

$S(f) = S_1(f) + S_2(f) + S_3(f)$



b) $m(t) = 2\cos(20\pi t) + 5\sin(40\pi t) \quad , f_1 = 10 \ , f_2 = 20$

$s(t) = [1 + k_a m(t)] \cos(2000\pi t) \quad , f_c = 1000$

$$s(t) = [1 + k_a[2\cos(20\pi t) + 5\sin(40\pi t)]]\cos(2000\pi t)$$
$$= \cos(2000\pi t) + k_a 2\cos(20\pi t)\cos(2000\pi t) + \overset{\frac{5}{2}\cdot 2}{5} k_a \sin(40\pi t)\cos(2000\pi t)$$
$$= \cos(2000\pi t) + k_a[\underbrace{\cos(20\pi t - 2000\pi t)}_{1} + \underbrace{\cos(20\pi t + 2000\pi t)}_{2}] + \frac{5}{2}k_a[\underbrace{\sin(40\pi t - 2000\pi t)}_{3} + \underbrace{\sin(40\pi t + 2000\pi t)}_{4}]$$

$$P_{SB} = \Sigma \frac{A^2}{2} = \cancel{2}\left(\frac{k_a^2}{\cancel{2}}\right) + \cancel{2}\left(\frac{(2.5k_a)^2}{\cancel{2}}\right) = k_a^2 + \frac{25}{4}k_a^2 = k_a^2\left(1 + \frac{25}{4}\right) = \frac{29}{4}k_a^2$$

$$P_{Total} = P_{carrier} + P_{SB} = \frac{1^2}{2} + P_{SB} = 1 + \frac{29}{4}k_a^2$$

Fourier Transforms

$\mathrm{rect}(t) \rightleftharpoons \mathrm{sinc}(f)$

$\mathrm{triang}(t) \rightleftharpoons \mathrm{sinc}^2(f)$

$\delta(t) \rightleftharpoons 1$

If $g(t) \rightleftharpoons G(f)$ and $h(t) \rightleftharpoons H(f)$, then

$$g(\overset{a}{t}) \rightleftharpoons \frac{1}{|a|} G(f/a)$$

$G(t) \rightleftharpoons g(-f)$

$g(t - t_0) \rightleftharpoons G(f)\exp(-j2\pi f t_0)$

$g(t)\exp(j2\pi f_c t) \rightleftharpoons G(f - f_c)$

$$g(t)\cos(2\pi f_c t) \rightleftharpoons \frac{1}{2}[G(f - f_c) + G(f + f_c)]$$

$$g(t)\sin(2\pi f_c t) \rightleftharpoons \frac{1}{2j}[G(f - f_c) - G(f + f_c)]$$

$g(t) * h(t) \rightleftharpoons G(f)H(f)$

$g(t)h(t) \rightleftharpoons G(f) * H(f)$

Filters :

$x(t) * h(t) = y(t)$

$X(f)H(f) = Y(f)$

AM :

standard AM: $s(t) = A_c[1 + k_a m(t)]\cos(2\pi f_c t)$

DSB-SC: $s(t) = A_c m(t)\cos(2\pi f_c t)$

Trigonometric Identities :

$2\cos(A)\cos(B) = \cos(A - B) + \cos(A + B)$

Set A=B, to find $\cos^2(A)$.

$2\sin A \sin B = \cos(A - B) - \cos(A + B)$

$2\sin A \cos B = \sin(A - B) + \sin(A + B)$

Mean Power of a sinusoid :

If a signal is of the $A\sin(2\pi f t + \theta)$, the mean power of the signal is $A^2/2$. The mean power does not depend on θ.

MIDTERM SOLUTIONS 2021

1. $x(t) = 3\,\delta(t+2) + 2\,\text{rect}(5t)$

$$\delta(t) \rightleftharpoons 1$$

$$\delta(t+2) \rightleftharpoons 1 \cdot e^{-j2\pi f(-2)} \qquad (\because\ \delta(t+2) = \delta(t-(-2)))$$

$$3\,\delta(t+2) \rightleftharpoons 3\,e^{+j4\pi f}$$

$$\text{rect}(t) \rightleftharpoons \text{sinc}(f)$$

SCALING PROPERTY: $\text{rect}(5t) \rightleftharpoons \frac{1}{5}\,\text{sinc}(f/5)$

$$2\,\text{rect}(5t) \rightleftharpoons \frac{2}{5}\,\text{sinc}(f/5)$$

LET $x(t) \rightleftharpoons X(f)$

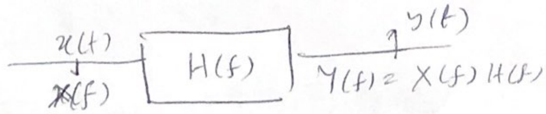
$$\therefore\ X(f) = 3\,e^{j4\pi f} + \frac{2}{5}\,\text{sinc}(f/5)$$

$$h(t) \rightleftharpoons H(f)$$

$$\text{rect}(5t) \rightleftharpoons \frac{1}{5}\,\text{sinc}(f/5)$$

$$\text{rect}([5(t-2)]) \rightleftharpoons \frac{1}{5}\,\text{sinc}(f/5) \cdot e^{-j2\pi f \cdot 2}$$

$$\therefore\ H(f) = \frac{4}{5}\,\text{sinc}(f/5) \cdot e^{-j4\pi f}$$



$$Y(f) = X(f)\,H(f)$$

$$= \left[3\,e^{j4\pi f} + \frac{2}{5}\,\text{sinc}(f/5)\right] \frac{4}{5} \cdot \text{sinc}(f/5)\,e^{-j4\pi f}$$

(2)

$$Y(f) = \underbrace{\frac{12}{5}\,\mathrm{SINC}(f/5)}_{Y_1(f)} + \underbrace{\frac{8}{25}\,\mathrm{SINC}^2(f/5)\, e^{-j2\pi f\cdot 2}}_{Y_2(f)}$$

$\underline{Y_1(f):}$ ~~$Y_1(f) = \frac{12}{25}$~~

CONSIDER

$$\mathrm{rect}(5t) \rightleftharpoons \frac{1}{5}\,\mathrm{SINC}(f/5)$$

$$y_1(t) = 12\,\mathrm{rect}(5t) \rightleftharpoons \frac{12}{5}\,\mathrm{SINC}(f/5) = Y_1(f)$$

$\underline{Y_2(f)}$:

CONSIDER

$$\mathrm{TRIANG}(t) \rightleftharpoons \mathrm{SINC}^2(f)$$

$$\mathrm{TRIANG}(5t) \rightleftharpoons \frac{1}{5}\,\mathrm{SINC}^2(f/5)$$

$$\frac{8}{5}\,\mathrm{TRIANG}(5t) \rightleftharpoons \frac{8}{25}\,\mathrm{SINC}^2(f/5)$$

USING TIME-SHIFTING PROPERTY

$$\frac{8}{5}\,\mathrm{TRIANG}[5(t-2)] \rightleftharpoons \frac{8}{25}\,\mathrm{SINC}^2(f/5)\, e^{-j2\pi f\cdot 2}$$

$$= Y_2(f)$$

$$\therefore\ y_2(t) = \frac{8}{5}\,\mathrm{TRIANG}[5(t-2)]$$

$$y(t) = y_1(t) + y_2(t)$$

$$= 12\,\mathrm{rect}(5t) + \frac{8}{5}\,\mathrm{TRIANG}[5(t-2)]$$

2. LET $m_1(t) = 2\cos(20\pi t)$

$m_2(t) = 5\sin(40\pi t)$

$m(t) = m_1(t) + m_2(t)$

Let $s_1(t) = k_a m_1(t)\cos(2000\pi t)$

$= 2k_a \cos(20\pi t)\cos(2000\pi t)$

$= \frac{\cancel{2}}{\cancel{2}} k_a\left[\cos[2020\pi t) + \cos(1980\pi t)\right]$

Let $s_2(t) = k_a m_2(t)\cos(2000\pi t)$

$= k_a \cdot 5 \underset{A}{\sin(40\pi t)} \underset{B}{\cos(2000\pi t)}$

$= \frac{5k_a}{2}\left[-\sin(1960\pi t) + \sin(2040\pi t)\right]$

$$s(t) = \overset{\mathrm{I}}{\cos(2000\pi t)} + k_a\left[\cos(2\pi\cdot 1010 t) + \overset{\mathrm{II}}{\cos(2\pi\cdot 990 t)}\right] + \frac{5k_a}{2}\left[-\underset{\mathrm{III}}{\sin(2\pi\cdot 980 t)} + \sin(1020\cdot 2\pi t)\right] \longrightarrow (*)$$

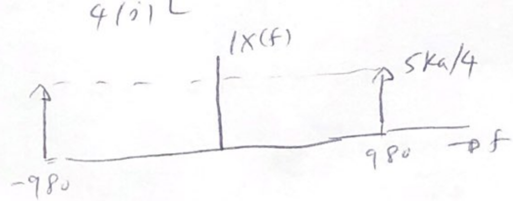
$\cos(2\pi f_0 t) \rightleftharpoons \frac{1}{2}\left[\delta(f-f_0) + \delta(f+f_0)\right]$

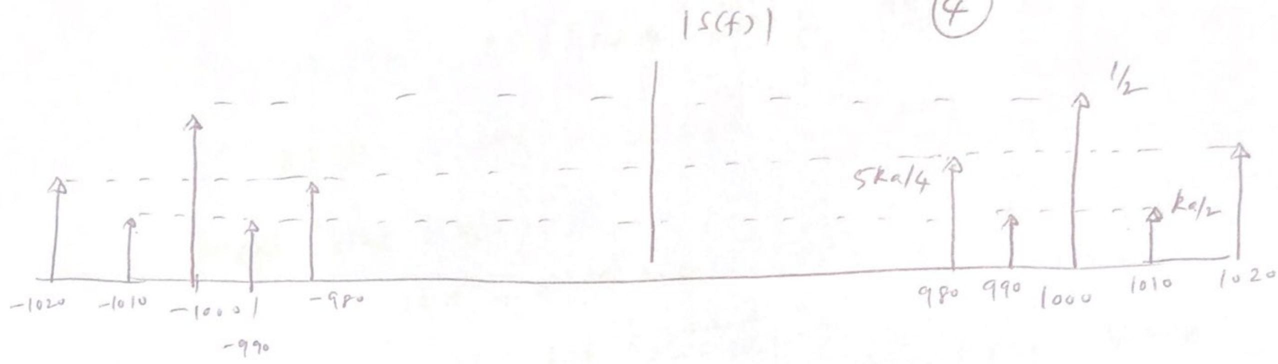
$\sin(2\pi f_0 t) \rightleftharpoons \frac{1}{2j}\left[\delta(f-f_0) - \delta(f+f_0)\right]$

LET $x(t) = -\frac{5k_a}{2}\sin(2\pi\cdot 980 t)$

$X(f) = -\frac{5k_a}{4j}\left[\delta(f-980) - \delta(f+980)\right]$

$|X(f)| = \frac{5k_a}{4|j|}\left[\delta(f-980) + \delta(f+980)\right]$ $|j| = 1$





THE MEAN POWER OF A SINUSOID, $A \cos(2\pi f_c t + \theta) = A^2/2$

$\therefore$ MEAN POWER OF THE CARRIER $= 1/2$ (SEE I OF ⊛)

MEAN POWER OF SIDEBANDS AT 1010 & 990 $\Big\} = \frac{1}{2} k_a^2 + \frac{1}{2} k_a^2$ (SEE II OF ⊛)

$= k_a^2$

MEAN POWER OF SIDEBANDS AT 980 & 1020 $\Big\} = \frac{1}{2}\left(\frac{5k_a}{2}\right)^2 + \frac{1}{2}\left(\frac{5k_a}{2}\right)^2$ (SEE III OF ⊛)

$= \frac{25k_a^2}{4}$

Total sideband power $= \left(1 + 25/4\right)k_a^2 = (29/4)k_a^2$

RATIO $= \dfrac{(29/4)k_a^2}{1 + 29/4\, k_a^2}$

Calculation of k_a:

To find the max or min of $m(t)$, differentiate it w.r.t. t:

$$\frac{dm}{dt} = 0 \Rightarrow -20\pi \cdot 2 \sin(20\pi t) + 5 \cdot 40\pi \cdot \cos(40\pi t) = 0$$

$$\Rightarrow 10 \cos(\underbrace{40\pi t}_{2\theta}) = 2 \sin(\underbrace{20\pi t}_{\theta})$$

5

$\cos(2\theta) = 1 - 2\sin^2\theta$

$\therefore \cancel{10}^{5} (1 - 2\sin^2\theta) = \cancel{2} \sin\theta$

Let $\sin\theta = x$

$10x^2 + x - 5 = 0$

$x = \dfrac{-2 \pm \sqrt{2 - 4 \cdot 10 \cdot (-5)}}{2} = 0.658$ or -0.758

$\theta = \sin^{-1} x = \underline{0.719}$ or $\underline{-0.861}$ rad

THE MAX or MIN OF $m(t)$ IS $2\cos\theta + 5\sin\theta$

$= 2\cos(0.719) + 5\sin(0.719)$

or

$2\cos(-0.861) + 5\sin(-0.861)$

$= 6.46$ or -3.63

$\therefore$ MAX $|m(t)| = 6.46$

$k_a \max|m(t)| \times 100 = 40$

$\Rightarrow k_a = \dfrac{40}{100 \times 6.46} = 0.0619$

$\text{RATIO} = \dfrac{(29/4)k_a^2}{(29/4)k_a^2 + 1/2} = 0.052$

5

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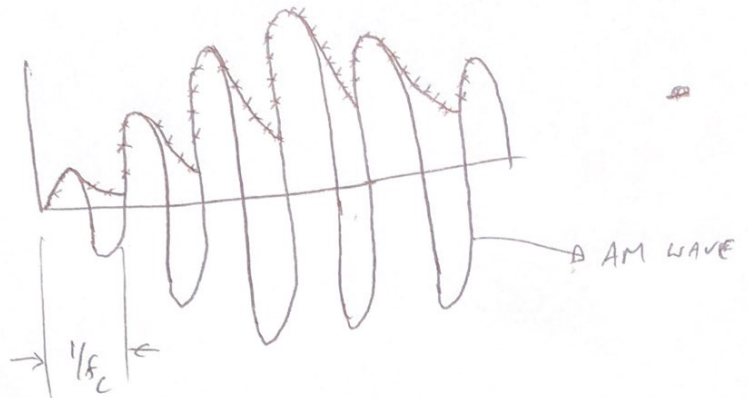
3. PRINCIPLE OF OPERATION → REFER TO LECTURE NOTES 'LEC4_AM'

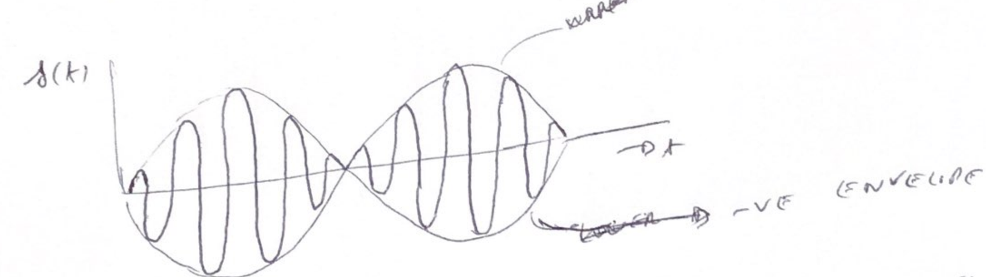
(a) $R_L C = 1/f_c$. IN THIS CASE, THE CAPACITOR DISCHARGES QUICKLY & THE RISE TIME/FALL TIME IS $\sim 1/f_c$.



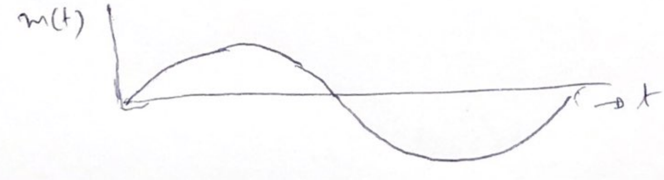
(b)

DSB-SC SIGNAL

$s(t) = A_c m(t) \cos(2\pi f_c t)$



AFTER PASSING THROUGH AN IDEAL ENVELOPE DETECTOR, IT DETECTS ONLY THE +VE SIDE OF ENVELOPE



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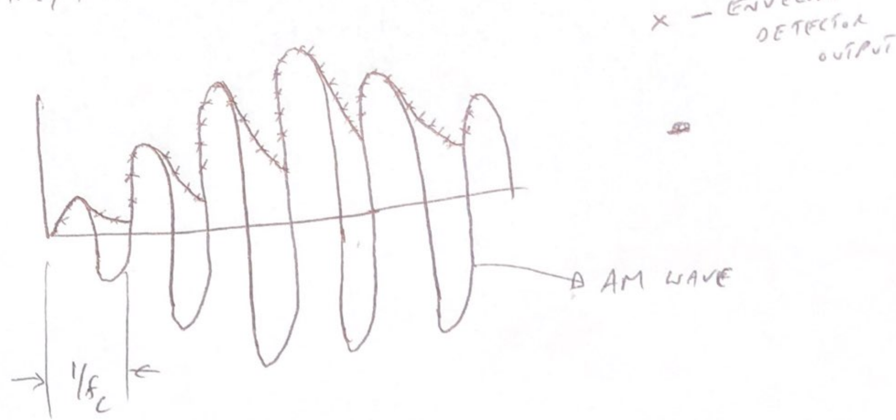

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DSB-SC SIGNAL

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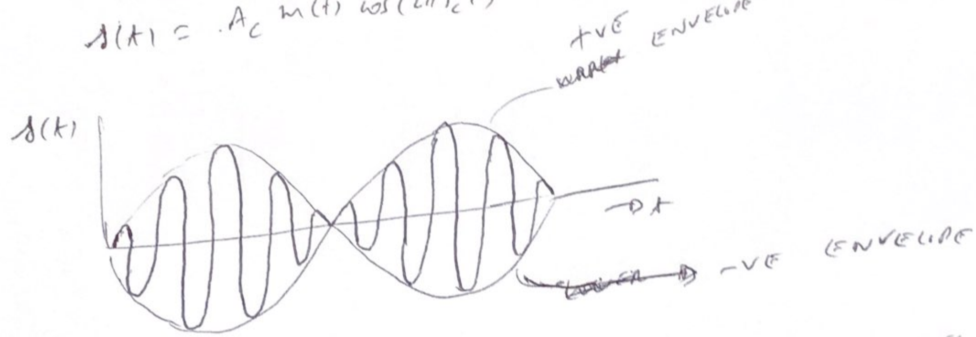

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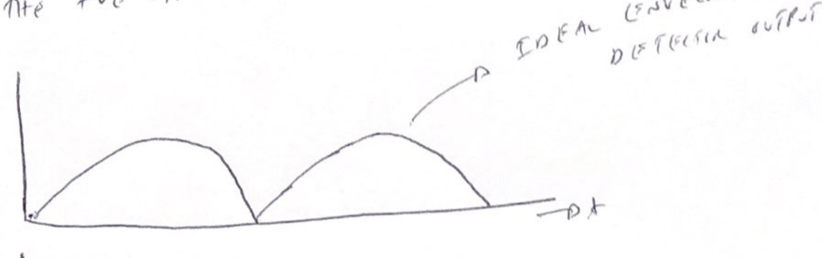

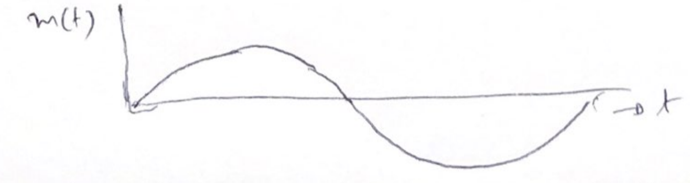

IT CAN BE SEEN THAT THERE IS DISTORTION IN THE -VE HALF CYCLE. ~~[illegible]~~ HENCE, MESSAGE SIGNAL CANNOT BE RETRIEVED. THE ENVELOPE DETECTOR OUTPUT IS ALWAYS +VE.